Technical Report: Exact Maximum Flexi-Clique Computation under Non-Hereditary Property and Connectivity

This report accompanies the paper. Section A gives the complete proofs of the two hardness results of the paper (Theorems 1 and 2 below), which share one reduction; Section B gives the complete proof of the exactness of EBA; Section C reports two additional experiments: the solution quality of the two heuristics, NPA and FPA, on the same instances (Section C.1), and the effect of the margin pivot on the branching width (Section C.2).

A Hardness of the maximum Flexi−clique problem

Fix a rational $\tau = p/q \in (0, 1)$, where $0 < p < q$ are coprime. A Flexi−clique is a node set H such that $G[H]$ is connected and every node of H has at least $\lfloor |H|^\tau \rfloor$ neighbours in H. For this fixed τ, the decision problem asks, given a graph G and an integer s, whether a Flexi−clique of size at least s exists.

THEOREM 1 (THEOREM 1 OF THE PAPER). *For every fixed rational $\tau \in (0, 1)$, the decision version of the maximum Flexi−clique problem is* NP*-complete.*

THEOREM 2 (THEOREM 2 OF THE PAPER). *For every fixed rational $\tau \in (0, 1)$ and every constant $\rho \geq 1$, it is* NP*-hard to approximate the maximum Flexi−clique problem within a factor of ρ.*

The earlier clique-based hardness argument lets τ depend on the input [4]; here τ is fixed, as it is in practice. Moreover, the approximation claim needs a gap reduction: it does not follow from decision hardness alone, nor does the inapproximability of the maximum clique problem carry over for a fixed τ, since the threshold $\lfloor |H|^\tau \rfloor$ falls far below the clique threshold $|H| - 1$ as $|H|$ grows.

Proof idea. We transform an instance (G, k) of the clique decision problem [3] into a graph $\widehat{G}$ and an integer U with the following gap:

$$\begin{aligned} G \text{ has a } k\text{-clique}: &\quad \mathrm{OPT}(\widehat{G}) \geq U, \\ G \text{ has no } k\text{-clique}: &\quad \mathrm{OPT}(\widehat{G}) < U/\rho. \end{aligned}$$

The graph $\widehat{G}$ consists of a large regular padding graph and the input graph G. We choose U at the upper end of a size interval on which the degree threshold equals an integer t. One padding node, the deficient node v_0, is missing k neighbours. At size U, exactly k nodes of G must repair this deficit, and their own degree requirements then force them to form a clique.

The remaining difficulty is to exclude solutions of intermediate size. The padding graph has its edges spread throughout its nodes. Keeping only a fraction σ of those nodes retains roughly a fraction σ of each degree, whereas the required degree scales as σ^τ. Since $\sigma^\tau > \sigma$ for $0 < \sigma < 1$, sufficiently large proper parts of the padding graph cannot meet the degree requirement. The proof below makes this comparison uniform, including sizes very close to the threshold interval. The circulant with consecutive offsets $1, \ldots, t/2$ would not do: a stretch of about $(t/2)^{1/\tau}$ consecutive residues is itself a Flexi−clique, because only the nodes near its ends lose neighbours.

A.1 The two properties of the padding graph

We need a regular graph with many edges across every cut and with few neighbours per node in a small attachment set. We obtain it from a circulant graph. For $S \subseteq \mathbb{Z}_N \setminus \{0\}$ with $S = -S$, the graph $C_N(S)$ has node set $\mathbb{Z}_N$ and edges $\{v, v + s\}$

for $v \in \mathbb{Z}_N$, $s \in S$. It is simple and $|S|$-regular, and its adjacency eigenvalues are

$$\mu_j = \sum_{s \in S} \cos(2\pi j s/N), \qquad j \in \mathbb{Z}_N,$$

with $\mu_0 = |S|$. Write λ for an upper bound on $|\mu_j|$ over all $j \neq 0$.

For disjoint node sets X, Y, let $e(X, Y)$ be the number of edges between them. The cut form of the expander mixing lemma [2] gives, in any t-regular graph R_0 on N nodes whose eigenvalues on the space orthogonal to the all-ones vector are at most λ in absolute value,

$$e_{R_0}(X, \overline{X}) \geq (t - \lambda) \frac{|X|\,(N - |X|)}{N}, \qquad \overline{X} = V(R_0) \setminus X. \tag{1}$$

This is the only mixing inequality used in the reduction; its short proof is in Section A.3.

LEMMA 1 (SPREAD-OUT CIRCULANT). *Let $N \geq 6$ and let t be even with $2 \leq t \leq (N-2)/2$. In time polynomial in N, one can construct a set $S \subseteq \mathbb{Z}_N \setminus \{0\}$ with $S = -S$, $|S| = t$, and $|\mu_j| \leq \lambda_N(t)$ for every $j \neq 0$, where*

$$\lambda_N(t) = 2\sqrt{t \ln(6N) + \frac{t^3}{N-2}} + \frac{2t}{N-2}. \tag{2}$$

LEMMA 2 (NEIGHBOURS IN AN INTERVAL). *In $C_N(S)$, if $|S| = t$ and $|\mu_j| \leq \lambda$ for every $j \neq 0$, then every node has at most*

$$\frac{t\ell}{N} + \lambda(1 + \ln N) \tag{3}$$

neighbours in any interval of ℓ consecutive residues.

Lemma 1 supplies the regular graph; Lemma 2 lets us choose a small attachment set without concentrating too many of any node's neighbours in that set. The supporting proofs are deferred to Section A.3.

A.2 The reduction

PROOF OF THEOREMS 1 AND 2. Membership in NP follows by checking connectivity and degrees: for an integer d, the test $\lfloor s^\tau \rfloor \leq d$ is equivalent to the integer comparison $s^p < (d+1)^q$.

Fix $\rho \geq 1$ ($\rho = 1$ for Theorem 1), put $\sigma_0 = 1/(3\rho)$, and let (G, k) be a clique instance with $n = |V(G)|$ and $3 \leq k \leq n$. Instances with smaller k are decided directly.

Parameters. Choose a sufficiently large even integer t, and set

$$L = \lceil t^{1/\tau} \rceil, \qquad U = \lceil (t+1)^{1/\tau} \rceil - 1, \qquad N = U - k. \tag{4}$$

Thus $\lfloor x^\tau \rfloor = t$ exactly when $L \leq x \leq U$. These integers are computed by integer comparisons with p and q; no real power has to be evaluated. Let

$$\lambda = \lambda_N(t), \qquad \Delta = k + 1 + \frac{t^2}{N} + \lambda(1 + \ln N). \tag{5}$$

Here Δ will bound the number of neighbours in the attachment set. Take

$$c = \tfrac{1}{2} \min\{1,\ 1/\tau - 1\}, \qquad t = \text{the smallest even integer at least } \max\{t_0, n^{1/c}\},$$

where t_0 depends only on τ and ρ and is chosen sufficiently large. Section A.3.1 verifies the following estimates, uniformly for $3 \le k \le n \le t^c$:

$$N \sim t^{1/\tau}, \qquad n = o(t), \qquad \lambda = o(t), \qquad \Delta = o(t), \tag{6}$$

$$N^\tau = t + 1 - o(1), \qquad t\Big(1 - \frac{t^{1/\tau}}{N}\Big) \longrightarrow \frac{1}{\tau}, \qquad \frac{N}{t(k+1)} \longrightarrow \infty. \tag{7}$$

Consequently one constant $t_0 = t_0(\tau, \rho)$ makes all of the following hold for every input instance: $N \ge 6$ and $t \le (N-2)/2$ (so Lemma 1 applies); $U \ge L$; $\lfloor N/(t-2k) \rfloor > k+1$; $t - k - \Delta > 0$; $\Delta < (t-\lambda)/2$; $t + n \le \sigma_0 N$; and the two inequalities (10) and (13) below. The resulting t, N and U are polynomial in n for fixed τ and ρ.

Construction. Construct $R_0 = C_N(S)$ with Lemma 1. Let v_0 denote the residue 0; it will be the deficient node, and its neighbours in R_0 are exactly the offsets $s \in S$. Choose distinct $s_1, \dots, s_k \in S$ and delete the k edges $\{v_0, s_i\}$, obtaining R. Choose an interval I of $t - 2k$ residues that avoids $v_0, s_1, \dots, s_k$, and put

$$A = \{v_0, s_1, \dots, s_k\} \cup I, \qquad B = \mathbb{Z}_N \setminus A.$$

Such an interval exists: among the $\lfloor N/(t-2k) \rfloor > k+1$ disjoint intervals of $t-2k$ residues, at most $k+1$ meet a forbidden node. In particular, $|A| = t - k + 1$. Form $\widehat{G}$ from disjoint copies of G and R by adding all edges between A and $V(G)$; it has $N + n$ nodes, and the output is $(\widehat{G}, U)$. The padding degrees in R are

$$d_R(v_0) = t - k, \qquad d_R(s_i) = t - 1, \qquad d_R(v) = t \quad (v \notin \{v_0, s_1, \dots, s_k\}).$$

Every node of B therefore has degree exactly t in $\widehat{G}$: it has no neighbour in G.

Two connectivity facts. By Lemma 2, every node of R_0 has at most

$$k + 1 + \frac{t(t-2k)}{N} + \lambda(1 + \ln N) \le \Delta$$

neighbours in A. Hence:

(i) Every node of A has a neighbour in B, because its degree in R is at least $t - k$ and $t - k - \Delta > 0$.

(ii) $R[B] = \widehat{G}[B]$ is connected. Otherwise choose a component W of $R[B]$ with $0 < |W| \le |B|/2 \le N/2$. All edges of R_0 leaving W end in A, since the deleted edges do not meet B. But (1) gives

$$\frac{t-\lambda}{2}\,|W| \le e_{R_0}(W, \overline{W}) \le \Delta |W|,$$

contradicting $\Delta < (t-\lambda)/2$.

In particular, R is connected.

YES case. Suppose $K \subseteq V(G)$ is a k-clique. Then $H = \mathbb{Z}_N \cup K$ has size U and threshold t. The deficient node v_0 has $(t-k) + k = t$ neighbours in H, and every node of K has $|A| + k - 1 = t$. Each s_i has $t - 1 + k \ge t$ neighbours, the other nodes of A have $t + k$, and those of B have t. As R is connected and K is joined to A, H is a Flexi–clique of size U.

NO case. Suppose G has no k-clique, and let H be a Flexi–clique of $\widehat{G}$. We show $|H| < U/\rho$.

Case 1: H avoids B. Then $|H| \le |A| + n \le t + n \le \sigma_0 N$.

Case 2: $|H| > U$. The threshold is at least $t + 1$, so H contains no node of B, and Case 1 applies; in fact this size range is impossible.

Case 3: $L \le |H| \le U$ and $H \cap B \ne \emptyset$. The threshold is t. A node of $H \cap B$, having total degree t, must keep all its neighbours. Connectivity of $\widehat{G}[B]$ forces $B \subseteq H$; as every node of A has a neighbour in B, it also forces $A \subseteq H$. Thus

all N padding nodes belong to H. Let $K = H \cap V(G)$; then $|K| \le U - N = k$. Node v_0 needs k neighbours from K, so $|K| = k$. Every node of K has at most $|A| = t - k + 1$ padding neighbours and therefore needs at least $k - 1$ neighbours in K. Hence K is a k-clique, a contradiction.

Case 4: $|H| < L$ and $H \cap B \neq \emptyset$. Put

$$X = H \cap \mathbb{Z}_N, \quad x = |X|, \quad Y = \mathbb{Z}_N \setminus X, \quad \sigma = x/N, \quad \varphi = 1 - \sigma.$$

We prove $x < \sigma_0 N$ by comparing two bounds on the number of edges from $X \cap B$ to the omitted padding nodes Y.

Upper bound: feasibility limits the lost neighbours. Each $b \in X \cap B$ has degree t and no neighbour in G, so it can lose at most $t - \lfloor |H|^\tau \rfloor \le t - \lfloor x^\tau \rfloor$ neighbours. Consequently,

$$e_R(X \cap B, Y) \le x\big(t - \lfloor x^\tau \rfloor\big) < x\,(t + 1 - x^\tau). \tag{8}$$

Lower bound: the padding graph forces many lost neighbours. Inequality (1) gives $e_{R_0}(X, Y) \ge (t - \lambda)x\varphi$. No deleted edge meets B, so the edges from $X \cap B$ to Y are the same in R and R_0, and excluding the nodes of $X \cap A$ from the cut loses at most $\Delta|Y| = \Delta N\varphi$ edges. Thus

$$e_R(X \cap B, Y) \ge (t - \lambda)x\varphi - \Delta N\varphi. \tag{9}$$

The bounds are incompatible when $\sigma \ge \sigma_0$. Assume $\sigma \ge \sigma_0$. Since $x \le |H| < L$, we have $x < t^{1/\tau}$, and (7) ensures

$$\varphi \ge \frac{1}{2\tau t} \tag{10}$$

for $t \ge t_0$. The elementary inequality

$$\sigma^\tau \ge \sigma\big(1 + (1 - \tau)(1 - \sigma)\big) \qquad (0 < \sigma \le 1) \tag{11}$$

is the tangent-line lower bound at $u = 1$ for the convex function $u^{\tau-1}$. Combining it with $N^\tau = t + 1 - o(1)$ gives

$$t + 1 - x^\tau \le (t + 1)\varphi\big(1 - (1 - \tau)\sigma\big) + o(1), \tag{12}$$

with an error that is uniform in $\sigma \in [\sigma_0, 1]$. Divide (9) by x and subtract the bound in (12). The difference is at least

$$\varphi\Big((t + 1)(1 - \tau)\sigma - 1 - \lambda - \frac{\Delta}{\sigma}\Big) - o(1) \;\ge\; \varphi\,\frac{(1 - \tau)\sigma_0 t}{2} - o(1) \;\ge\; \frac{(1 - \tau)\sigma_0}{4\tau} - o(1) \;>\; 0. \tag{13}$$

The first inequality uses $\lambda, \Delta = o(t)$ and $\sigma \ge \sigma_0$ (the bracket is increasing in σ); the second uses (10). Thus the lower bound exceeds the upper bound, a contradiction. We conclude $x < \sigma_0 N$, and hence $|H| \le x + n < \sigma_0 N + n \le 2\sigma_0 N$.

The gap and polynomial size. All cases give $|H| < 2\sigma_0 N = \frac{2N}{3\rho} < \frac{U}{\rho}$ in a NO instance, whereas a YES instance has a Flexi−clique of size U. The construction is deterministic and polynomial in n. For $\rho = 1$, deciding whether a Flexi−clique of size at least U exists decides the clique instance, which proves Theorem 1. For general fixed ρ, a ρ-approximation algorithm returns a Flexi−clique of size at least U/ρ in a YES instance and cannot do so in a NO instance, which proves Theorem 2. □

REMARK 1. *The constant $t_0(\tau, \rho)$ is large, so the reduction is of theoretical interest only; the instances are nevertheless of polynomial size in n for fixed τ and ρ. No limit has to be evaluated by the reduction algorithm: t_0 is a fixed constant for the fixed problem parameters.*

A.3 Supporting details

The reduction uses the graph properties as lemmas. We now justify the uniform parameter estimates, the cut inequality, and the two lemmas.

A.3.1 Why one polynomial choice of t suffices. Let $a = 1/\tau - 1 > 0$ and $c = \min\{1, a\}/2$. Then $c < 1$ and $c < a$. Uniformly for $k \le n \le t^c$,

$$n = o(t), \qquad k + 1 = o(t^a), \qquad N = (t+1)^{1/\tau} - O(k+1) \sim t^{1/\tau}.$$

In particular $N/(t(k+1)) \to \infty$ and $N/t \to \infty$. From (2),

$$\frac{\lambda}{t} = O\Big(\sqrt{\frac{\ln t}{t}} + \sqrt{\frac{t}{N}} + \frac{1}{N}\Big), \qquad \frac{\lambda \ln N}{t} \longrightarrow 0.$$

Together with $k/t \to 0$ and $t/N \to 0$, this gives $\Delta/t \to 0$.

Write $T = (t+1)^{1/\tau}$ and $\delta = T - N$. Since $U = \lceil T \rceil - 1$, we have $k < \delta \le k + 1$. As $N = T(1 - \delta/T)$ and $(1-z)^\tau \ge 1 - z$ for $0 \le z \le 1$,

$$0 \le t + 1 - N^\tau \le \frac{(t+1)(k+1)}{T} = O(t^{c-a}) = o(1).$$

Also

$$N - t^{1/\tau} = \big((t+1)^{1/\tau} - t^{1/\tau}\big) - \delta \sim \frac{1}{\tau}\, t^a,$$

because the first difference is asymptotic to $(1/\tau)t^a$ and $\delta = o(t^a)$. Dividing by $N \sim t^{a+1}$ proves $t(1 - t^{1/\tau}/N) \to 1/\tau$.

Every vanishing error above is bounded by a function of t and the fixed τ only, using $k, n \le t^c$. Therefore a single sufficiently large constant $t_0(\tau, \rho)$ ensures all strict inequalities used in the reduction, for every input instance. Finally, $t = O(n^{1/c} + t_0)$ and $N = O(t^{1/\tau})$ are polynomial in n.

A.3.2 The cut inequality. Let M be the adjacency matrix of a t-regular graph on N nodes whose eigenvalues on the space orthogonal to $\mathbf{1}$ are at most λ in absolute value. For $x = |X|$, decompose the indicator vector as $\mathbf{1}_X = (x/N)\mathbf{1} + z$ with $z \perp \mathbf{1}$. Then $\|z\|^2 = x(1 - x/N)$ and

$$\mathbf{1}_X^{\mathsf{T}} M \mathbf{1}_X = \frac{tx^2}{N} + z^{\mathsf{T}} M z \le \frac{tx^2}{N} + \lambda x\Big(1 - \frac{x}{N}\Big).$$

The quadratic form counts each edge inside X twice, so subtracting it from the degree sum tx counts each cut edge once, which gives (1).

A.3.3 Deterministically choosing distinct offsets.

PROOF OF LEMMA 1. This is a scalar conditional-expectation construction in the spirit of Alon and Roichman [1] and of its derandomisation by Wigderson and Xiao [5]; we include the distinct-offset argument and the constants explicitly.

Step 1: simplicity and regularity. Put $D = \{1, \dots, \lfloor (N-1)/2 \rfloor\}$, $d = |D|$, and $m = t/2$. We select distinct $g_1, \dots, g_m \in D$ and set $S = \{\pm g_1, \dots, \pm g_m\}$. This gives exactly t distinct non-zero offsets, with no loops or repeated neighbours. For $j \ne 0$, put $Z_j(g) = 2\cos(2\pi jg/N)$; then $\mu_j = \sum_{i=1}^m Z_j(g_i)$. The sum of $Z_j(g)$ over D is -1 for odd N and $-1 - (-1)^j$ for even N, so $\mathbb{E}_{g \in D} Z_j(g) \in [-2/d, 0]$.

Step 2: all eigenvalues at once. Set

$$\alpha = \ln(6N) + \frac{t^2}{N-2}, \qquad \theta = \sqrt{\alpha/t}, \qquad b = \frac{2t}{N-2}.$$

After i choices, let $P_j = \sum_{h=1}^{i} Z_j(g_h)$ and define the pessimistic estimator

$$\Phi_i = \sum_{j=1}^{N-1} \sum_{\varepsilon\in\{-1,1\}} \exp(\varepsilon\theta P_j)\Big(\mathbb{E}_{g\in D}\exp(\varepsilon\theta Z_j(g))\Big)^{m-i}.$$

All terms are positive. Since $Z_j(g) \in [-2,2]$ with mean in $[-2/d, 0]$, Hoeffding's lemma bounds each exponential moment by $\exp(2\theta/d + 2\theta^2)$, and $2m/d = t/d \le b$; hence

$$\Phi_0 \le 2(N-1)\exp(\theta b + t\theta^2).$$

Step 3: choose only unused offsets. For a hypothetical next choice uniform on all of D, the average of the next estimator equals Φ_i exactly. Only $d - i$ offsets are still allowed, so positivity gives

$$\min_{g\in D\setminus\{g_1,\dots,g_i\}} \Phi_{i+1}(g) \le \frac{d}{d-i}\,\Phi_i.$$

Choose an unused offset whose estimator is within a factor $1 + 1/(4t)$ of this minimum. As $m \le d/2$,

$$\sum_{i=0}^{m-1} \ln\frac{d}{d-i} \le \frac{m(m-1)}{d} \le \frac{t^2}{2(N-2)}, \qquad m\ln\big(1+1/(4t)\big) \le \frac{1}{8}.$$

Therefore

$$\ln\Phi_m \le \ln(2(N-1)) + \theta b + t\theta^2 + \frac{t^2}{2(N-2)} + \frac{1}{8} \le \alpha + \theta b + t\theta^2 = 2\alpha + \theta b.$$

For each j, one term of Φ_m equals $\exp(\theta|\mu_j|)$, so

$$|\mu_j| \le \frac{\ln\Phi_m}{\theta} \le 2\sqrt{t\alpha} + b = \lambda_N(t).$$

Polynomial-time evaluation. At each of $m \le N$ steps, evaluate at most N candidate estimators, each with $O(N)$ terms; the exponential moments over D are precomputed. Approximate each positive estimator with relative error at most $1/(16t)$ and select the smallest approximation; the selected true value is then within $(1 + 1/(16t))/(1 - 1/(16t)) \le 1 + 1/(4t)$ of the true minimum. All exponents have magnitude polynomial in N (indeed $\theta t = O(N)$), so the elementary functions and positive sums can be evaluated to the required relative accuracy with polynomially many bits in polynomial time, without exact comparisons of transcendental quantities. □

A.3.4 The interval neighbour bound.

PROOF OF LEMMA 2. For a node v and an interval I of length ℓ, the neighbours of v in I correspond to $S \cap J$, where $J = I - v$ is another interval of length ℓ. Use the Fourier transform $\widehat{f}(j) = \sum_{x\in\mathbb{Z}_N} f(x)\exp(-2\pi\mathrm{i}jx/N)$. The zero-frequency term contributes $t\ell/N$, and $\widehat{\mathbf{1}_S}(j) = \mu_j$ because $S = -S$. Parseval's identity therefore gives

$$|S\cap J| \le \frac{t\ell}{N} + \frac{\lambda}{N}\sum_{j=1}^{N-1}\big|\widehat{\mathbf{1}_J}(j)\big|.$$

For an interval, the geometric-series formula yields

$$\big|\widehat{\mathbf{1}_J}(j)\big| = \left|\frac{\sin(\pi j\ell/N)}{\sin(\pi j/N)}\right| \le \frac{1}{\sin(\pi j/N)} \qquad (j \ne 0).$$

Pairing j with $N-j$ and using $\sin(\pi u) \geq 2u$ on $[0, 1/2]$,

$$\sum_{j=1}^{N-1} \frac{1}{\sin(\pi j/N)} \leq N \sum_{j=1}^{\lfloor N/2 \rfloor} \frac{1}{j} \leq N(1 + \ln N).$$

Substitution proves (3). □

B Exactness of EBA

This section gives the complete proof of the exactness theorem of the paper (Theorem 3 below). We use the notation of Section V of the paper: a search node is $B = (S, C^r, C^{un}, D)$ with scope $H_B = S \cup C^r \cup C^{un}$; B *represents* a connected node set T if $S \subseteq T \subseteq H_B$; F' is the incumbent, $\theta_{F'} = \lfloor (|F'| + 1)^\tau \rfloor$ its degree threshold, $d_a(v) = d(v, G[H_B])$ the adjusted degree, L_u the distance from a candidate u to the farthest node of S in $G[H_B]$ and $\widehat{L}_u \leq L_u$ the lower bound that EBA maintains, $n(k, L)$ the degree–diameter bound (Lemma 3 of the paper), $C_\theta(v)$ the connected component of the $\theta_{F'}$-core of $G[H_B]$ containing v, and $s_{\max}(v)$, $\bar{d}_{\max}(v)$, $\bar{d}_S(v)$ and $\bar{d}_{\text{margin}}(v) = \bar{d}_{\max}(v) - \bar{d}_S(v)$ the budget quantities of the margin pivot. A search node is *extension-infeasible* if no set it represents is a Flexi−clique larger than F'.

THEOREM 3 (THEOREM 3 OF THE PAPER). *Given a graph $G = (V, E)$ and a parameter $\tau \in (0, 1)$, EBA returns a maximum Flexi−clique of G.*

The proof has three parts. Part 1 recalls what CP-Branching guarantees without any pruning: every connected node set is represented along exactly one root-to-leaf path, also when a search node branches on a pivot-selected subset of its reachable candidates. Part 2 shows that every reduction of EBA removes only nodes, search nodes or siblings that cannot lead to a Flexi−clique larger than the incumbent. Part 3 combines the two: the path of a maximum Flexi−clique is never cut, so the search reaches it unless the incumbent has already attained its size.

B.1 Part 1: CP-Branching enumerates every connected set exactly once

We use two facts proved in the paper (Lemma 2 and Proposition 1 there). First, the *invariant*: at every search node, S, C^r, C^{un} and D are pairwise disjoint; if $S \neq \emptyset$ then $G[S]$ is connected, every node of C^r is adjacent to S, and no node of C^{un} is adjacent to S. Second, the *partition property*: when a search node B is expanded, every connected set represented by B is represented by exactly one child of B. We restate the partition property in the generality in which EBA uses it, because the pivot of the paper branches on an ordering of a *subset* of C^r.

LEMMA 3 (PARTITION PROPERTY). *Let $B = (S, C^r, C^{un}, D)$ be a search node and T a connected set represented by B.*

(1) If $S \neq \emptyset$ and B is expanded along an ordering $\langle v_1, \ldots, v_k \rangle$ of any nonempty subset of C^r, with the children $B_1, \ldots, B_{k+1}$ of Equation (1) of the paper, then exactly one child represents T, namely B_i for the smallest i with $v_i \notin T$, or B_{k+1} if $\{v_1, \ldots, v_k\} \subseteq T$.

(2) If $S = \emptyset$ (the root) and the children are $B_i = (\{v_i\}, C_i^r, C_i^{un}, D \cup \{v_{i+1}, \ldots, v_k\})$ for the ordering $\langle v_1, \ldots, v_k \rangle$ of C^{un}, so that the scope of B_i is $\{v_1, \ldots, v_i\}$ minus removed nodes, then exactly one child represents T, namely B_i for the last *node v_i of T in the ordering.*

PROOF. (1) Let i be as stated. $S_i = S \cup \{v_1, \ldots, v_{i-1}\} \subseteq T$ by the choice of i. The scope of B_i is $H_B \setminus \{v_i\}$ for $i \leq k$ and H_B for $i = k + 1$ (Equation (1) moves nodes between S, C^r and C^{un} but removes only v_i from the scope), and $v_i \notin T$, so $T \subseteq H_{B_i}$; hence B_i represents T. A child B_j with $j < i$ has $v_j \in T$ but $v_j \notin H_{B_j}$, and a child B_j with $j > i$ has $v_i \in S_j$ but $v_i \notin T$; neither represents T. The candidates of C^r outside the ordering stay in C_j^r for every child, so the argument does not

depend on the ordering covering all of C^r. (2) Let v_i be the last node of T in the ordering. Then $T \subseteq \{v_1, \ldots, v_i\} = H_{B_i}$ and $S_i = \{v_i\} \subseteq T$, so B_i represents T. A child B_j with $j > i$ has $S_j = \{v_j\} \not\subseteq T$, and a child B_j with $j < i$ excludes $v_i \in T$ from its scope. □

Applying the lemma along the search tree, every connected set T of G with $T \subseteq V'$ (V' the node set after the global pruning) is represented by exactly one search node at each depth of the search tree, as long as the search node representing it is expanded; these search nodes form a path from the root. A search node on this path whose partial set equals T need not be a leaf, and its descendants on the path may still represent T: reachable candidates outside T may remain, and the children that exclude them keep $S = T$. What matters is that the path is finite. The search terminates because $|H_B \setminus S|$ strictly decreases from a search node to each of its children: a child of a search node with $S \neq \emptyset$ removes a candidate from the scope or adds a candidate to S, and a child of the root has a non-empty S inside a scope contained in V'. Hence every path of the search tree has length at most $|V'|$.

B.2 Part 2: every reduction is sound

A reduction is *sound* if it never removes a node of a Flexi−clique X represented by the current search node with $|X| > |F'|$, never prunes a search node that represents such an X, and never skips a sibling that represents such an X. We go through the reductions of Algorithm 1 of the paper in the order in which they are applied. Throughout, X denotes a Flexi−clique represented by the current search node B with $|X| > |F'|$; every node of X has at least $\lfloor |X|^\tau \rfloor \geq \lfloor (|F'| + 1)^\tau \rfloor = \theta_{F'}$ neighbours in X, and $X \subseteq H_B$.

Rule 1 (degree) and its cascade. Rule 1 discards $v \in H_B$ if $d_a(v) < \theta_{F'}$. Since $X \subseteq H_B$, every $v \in X$ has $d_a(v) \geq d(v, G[X]) \geq \theta_{F'}$, so v is not discarded. Removing a discarded node from the scope lowers the adjusted degrees of its neighbours; the cascade re-applies the same test to the reduced scope $H' \subseteq H_B$, and the same inequality $d(v, G[H']) \geq d(v, G[X]) \geq \theta_{F'}$ holds for every $v \in X$ as long as $X \subseteq H'$, which is maintained inductively. The *followers* of an excluded candidate v_i are the nodes whose degree in $G[H_B \setminus \{v_i\}]$ falls below $\theta_{F'}$; they are the first step of the same cascade in the child's scope, so the same argument applies. If the cascade reaches a node of S, then no set represented by B contains a Flexi−clique larger than F', because such a set would have to contain S, so the pruning of the whole search node is sound. The *global* application before the search and the *re-application at the root* (Rules 1 and 3 on the remaining scope whenever the incumbent has grown) are the same tests at the root search node with the current incumbent, hence sound by the same argument and by the one for Rule 3 below.

Rule 2 (distance). Lemma 4 of the paper shows that if Rule 2 prunes a candidate u with the exact distance L_u, no set represented by B that contains u is a Flexi−clique larger than F'. EBA applies the rule with a lower bound $\widehat{L}_u \leq L_u$ (computed in ancestor scopes, which contain the current scope, so that distances can only have grown): $n(k, L)$ is non-decreasing in L, so $n(\theta_{F'}, \widehat{L}_u) \leq n(\theta_{F'}, L_u)$, and both conditions of Rule 2 with $\widehat{L}_u$ imply the corresponding conditions with L_u. Likewise $\widehat{L}_u = \infty$ implies $L_u = \infty$, i.e., u is disconnected from some node of S in $G[H_B]$, in which case no connected set represented by B contains u. Hence Rule 2 as implemented is sound.

Rule 3 (core components). Rule 3 discards v if the connected component $C_\theta(v)$ of v in the $\theta_{F'}$-core of $G[H_B]$ has at most $|F'|$ nodes. Let $v \in X$. Every node of X has at least $\lfloor |X|^\tau \rfloor \geq \theta_{F'}$ neighbours in $G[X] \subseteq G[H_B]$ and X is connected, so X lies in one connected component of the $\theta_{F'}$-core of $G[H_B]$, namely $C_\theta(v)$; hence $|C_\theta(v)| \geq |X| > |F'|$ and v is not discarded (Lemma 5 of the paper). The same holds whenever the rule is re-applied to a reduced scope with a larger incumbent, since the argument uses only $X \subseteq H_B$ and the current $\theta_{F'}$.

Scope test. A search node with $|H_B| \le |F'|$ represents no set larger than F', so discarding it is sound.

Saturation of a node of S. For $v \in X$ we have $|X| \le s_{\max}(v)$: indeed $|X| \le |H_B|$, and $d(v, G[X]) \ge \lfloor |X|^\tau \rfloor$ with $d(v, G[X]) \le d_a(v)$ gives $|X|^\tau < d_a(v) + 1$, i.e., $|X| \le \lceil (d_a(v)+1)^{1/\tau} \rceil - 1$. Since $s - 1 - \lfloor s^\tau \rfloor$ is non-decreasing in s for $\tau \in (0,1)$, the number of non-neighbours of v in X is at most $|X| - 1 - \lfloor |X|^\tau \rfloor \le \bar{d}_{\max}(v)$. Now let $v \in S$ be saturated at B. Under condition (i), $s_{\max}(v) < |S| \le |X|$, contradicting $|X| \le s_{\max}(v)$. Under condition (ii), $s_{\max}(v) = |S|$ forces $|X| = |S|$, so $X = S$, but S is not a Flexi−clique. Under condition (iii), $\bar{d}_S(v) > \bar{d}_{\max}(v)$, but $S \subseteq X$ gives $\bar{d}_X(v) \ge \bar{d}_S(v)$, contradicting the bound above. Hence no such X exists and pruning B is sound.

Saturation of a candidate. The implementation also removes a candidate $u \in C^r \cup C^{un}$ at a search node with $S \neq \emptyset$ if (a) u is a non-neighbour of some $z \in S$ with $\bar{d}_{\text{margin}}(z) \le 0$, or (b) $|S| - |N(u) \cap S| > \bar{d}_{\max}(u)$. Both are sound: if $u \in X$, then in case (a) z has at least $\bar{d}_S(z) + 1 > \bar{d}_{\max}(z)$ non-neighbours in X, and in case (b) u has at least $|S| - |N(u) \cap S| > \bar{d}_{\max}(u)$ non-neighbours in X, each contradicting the bound of the previous paragraph. Removals shrink the scope and the adjusted degrees, which can only lower $\bar{d}_{\max}$, so repeating the tests, interleaved with the cascade of Rule 1, stays sound.

Sibling skipping. Let B with $S \neq \emptyset$ be expanded along the pivot ordering, and suppose the sibling loop stops at the first child B_j whose partial set S_j is saturated with respect to H_B, skipping $B_j, B_{j+1}, \ldots$. By sibling monotonicity, $S_j \subseteq S_l$ for every $l \ge j$, and $H_{B_l} \subseteq H_B$. A set represented by B_l contains $S_l \supseteq S_j$ and lies in H_B; if it were a Flexi−clique X with $|X| > |F'|$, the saturated node $v \in S_j$ would violate one of the three bounds of the saturation paragraph, evaluated with $|H_B|$ and the adjusted degrees in H_B, which are at least those in H_{B_l} (so $s_{\max}$ with respect to H_B is at least $s_{\max}$ with respect to the actual scope of X, and the contradiction derived there persists). Hence every skipped sibling is extension-infeasible.

Incumbent update. The incumbent is replaced by S only if $|S| > |F'|$ and S passes the Flexi−clique test; $G[S]$ is connected by the invariant, so F' is a Flexi−clique at all times, and the initial incumbent, the output of FPA, is a Flexi−clique by construction (Section IV of the paper).

B.3 Part 3: the search reaches a maximum Flexi−clique

Proof of Theorem 3. Let F^* be a maximum Flexi−clique of G and suppose, for contradiction, that EBA terminates with $|F'| < |F^*|$. Then $|F'| < |F^*|$ held throughout the run, so at every moment F^* was a Flexi−clique larger than the incumbent. By Part 2, the global pruning did not remove any node of F^*, so $F^* \subseteq V'$ and the root represents F^*. By Lemma 3 applied inductively, as long as a search node B representing F^* is expanded along some ordering, exactly one child represents F^*. We show that every search node B that represents F^* and has $S \neq F^*$ is expanded and that the child representing F^* is visited.

By Part 2, no reduction applied at B removes a node of F^*, prunes B, or declares B invalid, and $|H_B| \ge |F^*| > |F'|$, so B passes the scope test. If $S = \emptyset$, B is the root and is expanded over all remaining nodes of C^{un}; by Lemma 3(2) the child starting from the last node of F^* in the root ordering represents F^*, and it is visited: the root loop skips only the children B_i with $i \le |F'|$, whose scope $H_{B_i} \subseteq \{v_1, \ldots, v_i\}$ has at most $|F'|$ nodes, and those whose starting node v_i has degree below $\theta_{F'}$ in H_{B_i}; such children are extension-infeasible by the scope test and by Rule 1, and the re-application of Rules 1 and 3 before a child removes no node of F^*. If $S \neq \emptyset$, then $F^* \setminus S \neq \emptyset$ and $G[F^*]$ is connected, so some node of $F^* \setminus S$ is adjacent to S; it lies in H_B and is adjacent to S, hence in C^r by the invariant. Thus $C^r \neq \emptyset$ and B is not a leaf; it is expanded along the pivot ordering $\langle v_1, \ldots, v_k \rangle$, a nonempty subset of C^r. By Lemma 3(1) exactly one child B_i represents F^*. The sibling loop visits $B_1, B_2, \ldots$ in order and stops at the first child whose partial set is saturated with respect to H_B; by the sibling-skipping paragraph of Part 2 every child from that point on is extension-infeasible, whereas B_i represents F^* with $|F^*| > |F'|$ and so is not. Hence the loop reaches B_i, and B_i is visited.

Following the unique child that represents F^* from the root, the search therefore visits a sequence of search nodes all representing F^*. The sequence is finite (Part 1), so it has a last search node, and this last search node has $S = F^*$: a search node representing F^* with $S \neq F^*$ is expanded and its child representing F^* is visited, as shown, so it is not the last one. Let B be the first search node of the sequence with $S = F^*$. Every recursive call of Algorithm 1 of the paper tests its partial set before any reduction, and at B we have $|S| = |F^*| > |F'|$ and $S = F^*$ a Flexi−clique, so the incumbent is updated to F^*, contradicting $|F'| < |F^*|$ at termination. Hence $|F'| = |F^*|$ at termination, and since F' is always a Flexi−clique (Part 2), EBA returns a maximum Flexi−clique. □

REMARK 2. *The proof uses the pivot only through Lemma 3(1), which holds for any ordering of any nonempty subset of C^r, and through the saturation bounds. The pivot selection rule of the paper and the fallback to the full set C^r therefore affect the running time but not the correctness of EBA.*

C Additional experiments

C.1 Solution quality of NPA and FPA

Table 1 lists the sizes of the solutions returned by the two heuristics on the datasets of the paper at $\tau \in \{0.5, 0.7, 0.9\}$. The paper compares their running times (EQ1) and the size of the FPA solution with the optimum (EQ3); this table adds the direct comparison of the two heuristics. On 20 of the 39 instances the two return solutions of the same size; FPA finds a larger solution on 18 instances, by a factor of up to 7 (FB at $\tau = 0.5$: 183 versus 26), and a smaller one on a single instance (CD at $\tau = 0.5$: 323 versus 350). Together with EQ1 of the paper, where FPA is faster than NPA on the largest graphs (CS, LJ and SK) and by three orders of magnitude on LU, this shows that the new heuristic improves on NPA in solution quality as well as in scalability. The improvement carries over to the exact search, which uses the heuristic solution as its incumbent: a larger incumbent raises the degree threshold of Rules 1 and 3, so more nodes are removed before branching.

Table 1. Solution sizes of NPA and FPA on the datasets of the paper at $\tau \in \{0.5, 0.7, 0.9\}$. Bold marks the larger of the two where they differ; an underlined FPA value equals the NPA value.

Alg.	τ	FB	GO	FW	CA	CD	RO	LU	DB	AM	CS	WI	LJ	SK
NPA	0.50	26	135	133	164	**350**	37	5	881	17	29,496	425	13,683	118,507
	0.70	37	39	38	77	93	18	2	246	12	4,843	628	1,897	11,105
	0.90	25	10	10	27	27	7	2	114	8	845	433	520	4,516
FPA	0.50	**183**	135	133	**168**	323	**63**	**6**	**899**	**38**	**29,583**	**1,443**	**13,688**	118,507
	0.70	**47**	39	38	77	93	18	**4**	**259**	**16**	**4,846**	628	1,897	**11,152**
	0.90	**29**	10	10	27	**31**	7	2	114	8	**1,511**	433	520	4,516

C.2 Effect of the margin pivot

EQ5 of the paper compares EBA with EBA−Pivot by running time. Table 2 gives the underlying measurements for every instance of the six EQ5 datasets on which EBA branches at least once: the number of search-tree nodes, the maximum number of children explored per internal node (the branching width), and the running time, with and without the pivot. EBA−Pivot branches on all reachable candidates in ascending order of adjusted degree and keeps Rules 1–3, the scope test, and the sibling skipping. The widths come from an instrumented build of the same binaries that counts the children explored per node; its search-tree node counts coincide with those of the timed runs.

Table 2. Effect of the margin pivot on the instances of the EQ5 datasets on which EBA branches (the other instances are certified at the root and are identical under both variants): number of search-tree nodes, maximum number of children explored per internal node (branching width), and running time, for EBA and for EBA without the pivot. The seed children of the root are not counted, except on CS at $\tau = 0.95$, where no node below the root branches and the width is that of the root (its 125 seed children, each pruned at once).

		EBA			EBA−Pivot		
Dataset	τ	# search nodes	Max w.	Time (s)	# search nodes	Max w.	Time (s)
FB	0.55	6,534	57	0.659	5,904	64	0.379
	0.70	339	26	0.035	433	30	0.031
GO	0.75	1,715	13	0.138	4,766	14	0.212
	0.80	173	13	0.032	364	13	0.032
	0.85	7,755	13	0.534	17,536	13	0.871
	0.90	93,445	12	6.617	628,890	13	31.08
	0.95	8,010	12	1.02	156,148	12	10.85
FW	0.65	188	30	0.049	281	34	0.049
	0.70	59	17	0.022	66	25	0.020
	0.75	1,046	13	0.159	799	13	0.088
	0.80	99	13	0.025	113	13	0.025
	0.85	63	13	0.025	63	13	0.023
	0.90	36,114	12	4.707	144,627	12	13.53
	0.95	8,269	10	2.694	86,475	10	12.08
CA	0.85	249	23	0.041	303	23	0.039
	0.90	8,407,951	23	679.1	7,265,942	27	398.6
	0.95	321,113	20	25.75	2,021,376	20	110
LU	0.50	123	4	0.169	133	4	0.166
	0.55	24	3	0.160	26	3	0.137
	0.65	9	3	0.226	8	3	0.218
	0.80	6	3	0.157	6	3	0.161
	0.85	6	3	0.160	6	3	0.159
	0.90	6	3	0.156	6	3	0.138
	0.95	6	3	0.156	6	3	0.150
CS	0.95	125	125	22.43	125	125	22.43

The pivot never increases the maximum width: it is lower with the pivot on 7 of the 25 instances and equal on the other 18. Without the pivot the search has 1.7× as many nodes in geometric mean. In running time, EBA−Pivot is 1.3× slower in geometric mean and up to 10× slower (GO at $\tau = 0.95$); it is faster by more than a quarter on three instances (FB at 0.55, FW at 0.75, CA at 0.90). The differences of a few milliseconds on the remaining sub-second instances are within measurement noise.

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